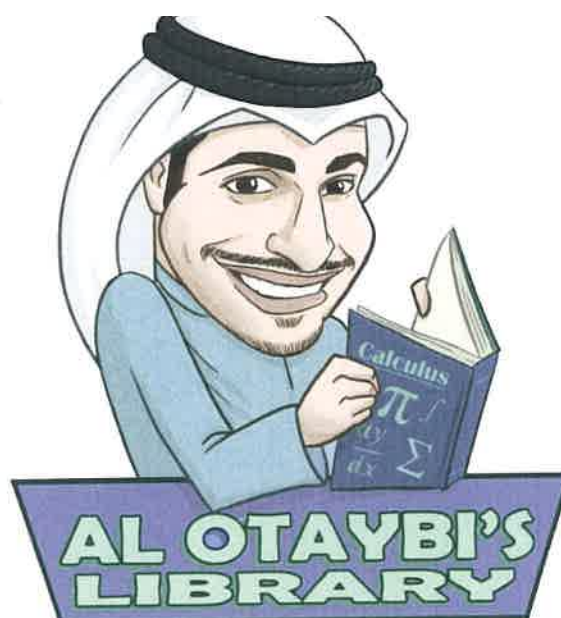


Kuwait University

2021/2022

Fall course



Calculus A

First

Sections : 2.6 – 2.7 – 2.8

2.6 Limits at Infinity ; Horizontal Asymptotes.

2.7 Derivatives and Rates of Change.

2.8 The Derivative as a Function.

(8)

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Good to know



$$\lim_{x \rightarrow \pm \infty}$$

$$\frac{3x^2 - 2x + 1}{5x^2 - 7}$$

$$= \frac{3}{5}$$

$$\frac{6x - 1}{9x^2 + 7}$$

$$= 0$$

$$\frac{4x^2 - 1}{x + 2}$$

$$= \pm \infty$$



التحليل الخاص



لو طلع من التعويض

$$\infty$$

$$-$$

$$\infty$$

هناك تحليل خاص

(1) أكبر عامل مشترك تحت الجذر (مع مراعاة في حال نتج عنها مطلق يجب إعادة تعريفه)



يمكنك تخمين الحل



(2) أكبر عامل مشترك بالبسط وبالمقام



Text book questions



Evaluate the limit if it exist

$$\lim_{x \rightarrow \infty} \frac{2x^2 - 7}{5x^2 + x - 3} \quad \frac{\infty}{\infty}$$

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\cancel{x^2} \left[2 - \frac{7}{x^2} \right]}{\cancel{x^2} \left[5 + \frac{1}{x^2} - \frac{3}{x^2} \right]} \\ = \frac{2 - \frac{7}{\infty}}{5 + \frac{1}{\infty} - \frac{3}{\infty}} \\ = \frac{2 - 0}{5 + 0 - 0} = \frac{2}{5} \end{aligned}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow \infty} \frac{3x + 5}{x - 4} \quad \frac{\infty}{\infty}$$

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\cancel{x} \left[3 + \frac{5}{x} \right]}{\cancel{x} \left[1 - \frac{4}{x} \right]} \\ = \frac{3 + \frac{5}{\infty}}{1 - \frac{4}{\infty}} \\ = \frac{3 + 0}{1 - 0} = \frac{3}{1} = 3 \end{aligned}$$

Evaluate the limit if it exist

$$\lim_{y \rightarrow \infty} \frac{2 - 3y^2}{5y^2 + 4y} \quad \frac{-\infty}{\infty}$$

$$\begin{aligned} \lim_{y \rightarrow \infty} \frac{\cancel{y^2} \left[\frac{2}{y^2} - 3 \right]}{\cancel{y^2} \left[5 + \frac{4}{y} \right]} \\ = \frac{\frac{2}{\infty} - 3}{5 + \frac{4}{\infty}} \\ = \frac{0 - 3}{5 + 0} = \frac{-3}{5} \end{aligned}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow -\infty} \frac{1 - x - x^2}{2x^2 - 7} \quad \frac{\infty}{\infty}$$

$$\begin{aligned} \lim_{x \rightarrow -\infty} \frac{\cancel{x^2} \left[\frac{1}{x^2} - \frac{1}{x} - 1 \right]}{\cancel{x^2} \left[2 - \frac{7}{x^2} \right]} \\ = \frac{\frac{1}{\infty} - \frac{1}{-\infty} - 1}{2 - \frac{7}{\infty}} \\ = \frac{0 - 0 - 1}{2 - 0} = \frac{-1}{2} \end{aligned}$$



Text book questions



Evaluate the limit if it exist

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{9x^6 - x}}{x^3 + 1} \quad \frac{\infty}{-\infty}$$

$$\begin{aligned} \lim_{x \rightarrow -\infty} \frac{\sqrt{x^6 (9 - \frac{1}{x^6})}}{x^3 + 1} &= \lim_{x \rightarrow -\infty} \frac{\sqrt{x^6} \sqrt{9 - \frac{1}{x^6}}}{x^3 + 1} \\ &= \lim_{x \rightarrow -\infty} \frac{x^3 \sqrt{9 - \frac{1}{x^6}}}{x^3 + 1} \\ &= \lim_{x \rightarrow -\infty} \frac{\cancel{x^3} \sqrt{9 - \frac{1}{x^6}}}{\cancel{x^3} [1 + \frac{1}{x^3}]} = \frac{\sqrt{9 - 0}}{1 + 0} = 3 \end{aligned}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow \infty} \frac{\sqrt{x + 3x^2}}{4x - 1} \quad \frac{\infty}{\infty}$$

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\sqrt{x^2 (\frac{1}{x} + 3)}}{4x - 1} &= \lim_{x \rightarrow \infty} \frac{\sqrt{x^2} \sqrt{\frac{1}{x} + 3}}{4x - 1} \\ \lim_{x \rightarrow \infty} \frac{1 \times 1 \sqrt{\frac{1}{x} + 3}}{4x - 1} \\ \lim_{x \rightarrow \infty} \frac{x \sqrt{\frac{1}{x} + 3}}{4x - 1} \\ \lim_{x \rightarrow \infty} \frac{\cancel{x} \sqrt{\frac{1}{x} + 3}}{\cancel{x} [4 - \frac{1}{x}]} &= \frac{\sqrt{0 + 3}}{4 - 0} = \frac{\sqrt{3}}{4} \end{aligned}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2 + 3x}}{2x + 1}$$

$$\frac{\infty}{\infty}$$

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 \left(4 + \frac{3}{x} \right)}}{2x + 1}$$

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2} \sqrt{4 + \frac{3}{x}}}{2x + 1}$$

$$\lim_{x \rightarrow -\infty} \frac{|x| \sqrt{4 + \frac{3}{x}}}{2x + 1}$$

$$\lim_{x \rightarrow -\infty} \frac{-x \sqrt{4 + \frac{3}{x}}}{2x + 1}$$

$$\lim_{x \rightarrow -\infty} \frac{-x \sqrt{4 + \frac{3}{x}}}{x \left[2 + \frac{1}{x} \right]}$$

$$= \frac{-\sqrt{4 + \frac{3}{-\infty}}}{2 + \frac{1}{-\infty}}$$

$$= \frac{-\sqrt{4}}{2} = \frac{-2}{2} = -1$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow \infty} (\sqrt{2x^2 + 3} - \sqrt{2x^2 - 5})$$

$\infty - \infty$

$$\lim_{x \rightarrow \infty} (\sqrt{2x^2 + 3} - \sqrt{2x^2 - 5}) \quad \frac{\sqrt{2x^2 + 3} + \sqrt{2x^2 - 5}}{\sqrt{2x^2 + 3} + \sqrt{2x^2 - 5}}$$

$$= \lim_{x \rightarrow \infty} \frac{2x^2 + 3 - [2x^2 - 5]}{\sqrt{2x^2 + 3} + \sqrt{2x^2 - 5}}$$

$$= \lim_{x \rightarrow \infty} \frac{\cancel{2x^2} + 3 - \cancel{2x^2} + 5}{\sqrt{2x^2 + 3} + \sqrt{2x^2 - 5}} = \frac{8}{\sqrt{\infty} + \sqrt{\infty}} = \frac{8}{\infty} = 0$$

Evaluate the limit if it exist

July 2017

$$\lim_{x \rightarrow \infty} (\sqrt{4x^2 + 1} - 2x)$$

$\infty - \infty$

$$\lim_{x \rightarrow \infty} (\sqrt{4x^2 + 1} - 2x) \quad \frac{\sqrt{4x^2 + 1} + 2x}{\sqrt{4x^2 + 1} + 2x}$$

$$= \lim_{x \rightarrow \infty} \frac{\cancel{4x^2} + 1 - \cancel{4x^2}}{\sqrt{4x^2 + 1} + 2x}$$

$$= \frac{1}{\sqrt{\infty} + \infty} = \frac{1}{\infty} = 0$$



Text book questions



Evaluate the limit if it exist

$$\lim_{x \rightarrow \infty} (\sqrt{9x^2 + x} - 3x)$$

$\infty - \infty$

$$\lim_{x \rightarrow \infty} (\sqrt{9x^2 + x} - 3x) \quad \frac{\sqrt{9x^2 + x} + 3x}{\sqrt{9x^2 + x} + 3x}$$

$$= \lim_{x \rightarrow \infty} \frac{\cancel{9x^2} + x - \cancel{9x^2}}{\sqrt{9x^2 + x} + 3x}$$

$$= \lim_{x \rightarrow \infty} \frac{x}{\sqrt{9x^2 + x} + 3x} \quad \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2(9 + \frac{1}{x})} + 3x}$$

$$= \lim_{x \rightarrow \infty} \frac{x}{|x| \sqrt{9 + \frac{1}{x}} + 3x}$$

$$= \lim_{x \rightarrow \infty} \frac{x}{x \sqrt{9 + \frac{1}{x}} + 3x}$$

$$= \lim_{x \rightarrow \infty} \frac{\cancel{x}}{\cancel{x} [\sqrt{9 + \frac{1}{x}} + 3]} = \frac{1}{\sqrt{9+0} + 3} = \frac{1}{3+3} = \frac{1}{6}$$



Text book questions



Evaluate the limit if it exist

$$\lim_{x \rightarrow -\infty} (\sqrt{4x^2 + 3x} + 2x)$$

$\infty - \infty$

$$\lim_{x \rightarrow -\infty} (\sqrt{4x^2 + 3x} + 2x) \quad \frac{\sqrt{4x^2 + 3x} - 2x}{\sqrt{4x^2 + 3x} - 2x}$$

$$\lim_{x \rightarrow -\infty} \frac{\cancel{4x^2} + 3x - \cancel{4x^2}}{\sqrt{4x^2 + 3x} - 2x} \quad \frac{-\infty}{\infty + \infty} \quad \frac{-\infty}{\infty}$$

$$\lim_{x \rightarrow -\infty} \frac{3x}{\sqrt{x^2(4 + \frac{3}{x})} - 2x} = \lim_{x \rightarrow -\infty} \frac{3x}{|x| \sqrt{4 + \frac{3}{x}} - 2x}$$

$$\lim_{x \rightarrow -\infty} \frac{3x}{-x \sqrt{4 + \frac{3}{x}} - 2x}$$

$$\lim_{x \rightarrow -\infty} \frac{3x}{-x \left[\sqrt{4 + \frac{3}{x}} - 2 \right]}$$

$$= \frac{3}{-\sqrt{4+0} - 2} = \frac{3}{-2-2} = -\frac{3}{4}$$



Text book questions



Evaluate the limit if it exist

$$\lim_{x \rightarrow \infty} \sqrt{\frac{9x^3 + 8x - 4}{3 - 5x + x^3}}$$

$$\sqrt{\frac{\infty}{\infty}}$$

$$\lim_{x \rightarrow \infty} \sqrt{\frac{\cancel{x^3} \left[9 + \frac{8}{x^2} - \frac{4}{x^3} \right]}{\cancel{x^3} \left[\frac{3}{x^3} - \frac{5}{x^2} - 1 \right]}}$$

$$= \sqrt{\frac{9 + 0 - 0}{0 - 0 - 1}} = \sqrt{9} = 3$$

Evaluate the limit if it exist

$$\lim_{t \rightarrow \infty} \frac{t - t\sqrt{t}}{2t^{3/2} + 3t - 5}$$

$$\frac{\infty}{\infty}$$

$$\lim_{t \rightarrow \infty} \frac{t - t^{\frac{3}{2}}}{2t^{\frac{3}{2}} + 3t - 5}$$

$$= \lim_{t \rightarrow \infty} \frac{\cancel{t^{\frac{3}{2}}} \left[\frac{t}{t^{\frac{3}{2}}} - 1 \right]}{\cancel{t^{\frac{3}{2}}} \left[2 + \frac{3t}{t^{\frac{3}{2}}} - \frac{5}{t^{\frac{3}{2}}} \right]}$$

$$= \lim_{t \rightarrow \infty} \frac{\frac{1}{t^{\frac{1}{2}}} - 1}{2 + \frac{3}{t^{\frac{1}{2}}} - \frac{5}{t^{\frac{3}{2}}}} = \frac{0 - 1}{2 + 0 - 0} = -\frac{1}{2}$$



Text book questions



Evaluate the limit if it exist

$$\lim_{x \rightarrow \infty} \frac{e^{3x} - e^{-3x}}{e^{3x} + e^{-3x}}$$

$$\frac{e^{\infty} - e^{-\infty}}{e^{\infty} + e^{-\infty}} = \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow \infty} \frac{\cancel{e^{3x}} \left[1 - \frac{e^{-3x}}{\cancel{e^{3x}}} \right]}{\cancel{e^{3x}} \left[1 + \frac{e^{-3x}}{\cancel{e^{3x}}} \right]}$$

$$\lim_{x \rightarrow \infty} \frac{1 - e^{-3x}}{1 + e^{-3x}} = \frac{1 - e^{-\infty}}{1 + e^{-\infty}} = \frac{1 - 0}{1 + 0} = 1$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow \infty} \frac{1 - e^x}{1 + 2e^x}$$

$$\frac{1 - e^{\infty}}{1 + 2e^{\infty}} = \frac{-\infty}{\infty}$$

$$\lim_{x \rightarrow \infty} \frac{\cancel{e^x} \left[\frac{1}{\cancel{e^x}} - 1 \right]}{\cancel{e^x} \left[\frac{1}{\cancel{e^x}} + 2 \right]}$$

$$\lim_{x \rightarrow \infty} \frac{e^{-x} - 1}{e^{-x} + 2} = \frac{\overset{0}{e^{-\infty}} - 1}{\underset{0}{e^{-\infty}} + 2} = \frac{-1}{2}$$



Text book questions



Evaluate the limit if it exist

$$\lim_{x \rightarrow \infty} e^{-2x} \cos x$$

$$e^{-\infty} \cos \infty$$

$$-e^{-2x} < \cos x < e^{-2x}$$

$\div e^{-2x} > 0$
* e^{-2x}

$$-e^{-2x} < e^{-2x} \cos x < e^{-2x}$$

$$\lim_{x \rightarrow \infty} -e^{-2x} = -e^{-\infty} = 0$$

$$\lim_{x \rightarrow \infty} e^{-2x} = e^{-\infty} = 0$$

$\therefore$ By Sandwich Theorem $\rightarrow \lim_{x \rightarrow \infty} e^{-2x} \cos x = 0$

Evaluate the limit if it exist

$$\lim_{x \rightarrow \infty} \frac{\sin^2 x}{x^2 + 1}$$

$$\frac{\sin^2 \infty}{\infty}$$

$$0 < \sin^2 x < 1$$

$\div (x^2 + 1) \because (x^2 + 1) > 0$

$$0 < \frac{\sin^2 x}{x^2 + 1} < \frac{1}{x^2 + 1}$$

$$\lim_{x \rightarrow \infty} (0) = 0$$

$$\lim_{x \rightarrow \infty} \frac{1}{x^2 + 1} = \frac{1}{\infty} = 0$$

$\therefore$ By Sandwich Theorem $\rightarrow \lim_{x \rightarrow \infty} \frac{\sin^2 x}{x^2 + 1} = 0$



Nice question !!



$$\lim_{x \rightarrow \infty} \frac{x \sin x}{x^2 + 1}$$

$$\frac{\infty \sin \infty}{\infty}$$

$$-1 \leq \sin \theta \leq 1 \quad \forall \theta$$

$$-1 \leq \sin x \leq 1$$

$$\text{as } x \rightarrow \infty \Rightarrow \frac{x}{x^2 + 1} \rightarrow 0$$

$$-\frac{x}{x^2 + 1} \leq \frac{x \sin x}{x^2 + 1} \leq \frac{x}{x^2 + 1}$$

$$\lim_{x \rightarrow \infty} \frac{-x}{x^2 + 1} = \frac{-\infty}{\infty}$$

$$\lim_{x \rightarrow \infty} \frac{-x}{x^2 (1 + \frac{1}{x^2})} = \lim_{x \rightarrow \infty} \frac{-1}{x (1 + \frac{1}{x^2})}$$

$$= \frac{-1}{\infty (1 + 0)} = 0$$

$$\lim_{x \rightarrow \infty} \frac{x}{x^2 + 1} = \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow \infty} \frac{x}{x^2 (1 + \frac{1}{x^2})} = \lim_{x \rightarrow \infty} \frac{1}{x (1 + \frac{1}{x^2})}$$

$$= \frac{1}{\infty (1 + 0)} = 0$$

$\Rightarrow \therefore$ By Sandwich
 $\lim_{x \rightarrow \infty} f(x) = 0$

Find $\lim_{x \rightarrow \infty} f(x) \quad \forall x > 1$ if

$$\frac{8e^x - 2}{2e^x} < f(x) < \sqrt{\frac{16x}{x-1}}$$

$$\lim_{x \rightarrow \infty} \frac{8e^x - 2}{2e^x} \quad \frac{8e^\infty - 2}{2e^\infty} \quad \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow \infty} \frac{\cancel{e^x} [8 - \frac{2}{e^x}]}{\cancel{2e^x}} = \frac{8 - \frac{2}{\infty}}{2} = \frac{8 - 0}{2} = 4$$

$$\lim_{x \rightarrow \infty} \sqrt{\frac{16x}{x-1}} = \lim_{x \rightarrow \infty} \frac{\sqrt{16x}}{\sqrt{x-1}} \quad \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow \infty} \frac{\sqrt{16x}}{\sqrt{x(1 - \frac{1}{x})}} = \lim_{x \rightarrow \infty} \frac{\sqrt{16} \sqrt{x}}{\sqrt{x} \sqrt{1 - \frac{1}{x}}} = \frac{4}{\sqrt{1 - \frac{1}{\infty}}} = \frac{4}{\sqrt{1}} = 4$$

∴ By Sandwich Theorem

$$\lim_{x \rightarrow \infty} f(x) = 4$$



Text book questions



Evaluate the limit if it exist

$$\lim_{x \rightarrow \infty} \arctan(e^x)$$

$$\lim_{x \rightarrow \infty} \tan^{-1}(e^x)$$

$$= \tan^{-1}(e^{\infty})$$

$$= \tan^{-1}(\infty) = \frac{\pi}{2}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0^+} \tan^{-1}(\ln x)$$

$$= \tan^{-1}(-\infty)$$

$$= -\frac{\pi}{2}$$



Asymptotes



Horizontal Asymptote

المحاذاة الأفقية

خطوات الحل

$$\lim_{x \rightarrow \pm \infty}$$

(1) أخذ اللمت لـ

(2) حل مسألة اللمت

يجب أن يكون ناتج اللمت

رقم

$$y = \text{رقم}$$

is Horizontal Asymptote

Vertical Asymptote

المحاذاة الرأسية

خطوات الحل

(1) استخراج أصفار المقام.

(2) أخذ اللمت لكل صفر بالاتجاهين.

يجب أن يكون ناتج اللمت

$$\pm \infty$$

$$x = \text{صفر المقام}$$

is Vertical Asymptote

Find the horizontal asymptote for

$$f(x) = \frac{3x^2 - 5x + 1}{2x^2 - 4}$$

$$\lim_{x \rightarrow \pm\infty} \frac{3x^2 - 5x + 1}{2x^2 - 4} \quad \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow \pm\infty} \frac{\cancel{x^2} \left[3 - \frac{5}{x} + \frac{1}{x} \right]}{\cancel{x^2} \left[2 - \frac{4}{x^2} \right]} = \frac{3 - 0 + 0}{2 - 0} = \frac{3}{2}$$

$\Rightarrow \therefore y = \frac{3}{2}$ is H.A

Find the horizontal asymptote for

$$f(x) = \frac{x - 4}{x^2 + 3x + 10}$$

$$\lim_{x \rightarrow \pm\infty} \frac{x - 4}{x^2 + 3x + 10} \quad \frac{\pm\infty}{\infty}$$

$$\lim_{x \rightarrow \pm\infty} \frac{\cancel{x} \left[1 - \frac{4}{x} \right]}{\cancel{x^2} \left[1 + \frac{3}{x} + \frac{10}{x^2} \right]} = \frac{1 - 0}{\pm\infty [1 + 0 + 0]} = \frac{1}{\pm\infty} = 0$$

$\Rightarrow \therefore y = 0$ is H.A

Find the horizontal asymptote for

$$f(x) = \frac{x^3 - 5x^2 + 4}{x^2 - 8}$$

$$\lim_{x \rightarrow \pm\infty} \frac{x^3 - 5x^2 + 4}{x^2 - 8} \quad \frac{\pm\infty}{\infty}$$

$$\lim_{x \rightarrow \pm\infty} \frac{\cancel{x^3} \left[1 - \frac{5}{x} + \frac{4}{x^3} \right]}{\cancel{x^2} \left[1 - \frac{8}{x^2} \right]} = \frac{\pm\infty [1 - 0 + 0]}{1 - 0} = \pm\infty$$

$\Rightarrow \therefore$ No horizontal asymptote.

Find the horizontal asymptote for

$$f(x) = \frac{\sqrt{x^2 + 4}}{2x + 4}$$

$$\lim_{x \rightarrow \pm\infty} \frac{\sqrt{x^2 + 4}}{2x + 4} \quad \frac{\infty}{\pm\infty}$$

$$\lim_{x \rightarrow \pm\infty} \frac{\sqrt{x^2 \left[1 + \frac{4}{x^2} \right]}}{2x + 4}$$

$$\lim_{x \rightarrow \pm\infty} \frac{|x| \sqrt{1 + \frac{4}{x^2}}}{2x + 4}$$

$$\lim_{x \rightarrow \infty} \frac{x \sqrt{1 + \frac{4}{x^2}}}{2x + 4}$$

$$\lim_{x \rightarrow -\infty} \frac{-x \sqrt{1 + \frac{4}{x^2}}}{2x + 4}$$

$$\lim_{x \rightarrow \infty} \frac{\cancel{x} \sqrt{1 + \frac{4}{x^2}}}{\cancel{x} \left[2 + \frac{4}{x} \right]}$$

$$\lim_{x \rightarrow -\infty} \frac{\cancel{-x} \sqrt{1 + \frac{4}{x^2}}}{\cancel{x} \left[2 + \frac{4}{x} \right]}$$

$$= \frac{\sqrt{1+0}}{2+0} = \frac{1}{2} \quad \bigg| \quad = \frac{-\sqrt{1+0}}{2+0} = -\frac{1}{2}$$

$\Rightarrow \therefore y = \frac{1}{2}$ and $y = -\frac{1}{2}$ are HA's.



أفترق $\pm\infty$ عن بعض في:

$$1 - \frac{e}{a} - \tan^{-1}$$



Text book question



Find the horizontal and vertical asymptotes (if any) of the following function

$$y = \frac{5 + 4x}{x + 3}$$

Horizontal Asymptote

$$\lim_{x \rightarrow \pm\infty}$$

$$\frac{5 + 4x}{x + 3}$$

$$\frac{\pm\infty}{\pm\infty}$$

$$\lim_{x \rightarrow \pm\infty}$$

$$\frac{\cancel{x} \left[\frac{5}{\cancel{x}} + 4 \right]}{\cancel{x} \left[1 + \frac{3}{\cancel{x}} \right]}$$

$$= \frac{\frac{5}{\pm\infty} + 4}{1 + \frac{3}{\pm\infty}} = \frac{4}{1} = 4$$

$\Rightarrow \therefore y = 4$ is H.A

Vertical Asymptote

$$x = -3$$

$$x + 3 = 0 \rightarrow x = -3$$

$$\lim_{x \rightarrow -3^+}$$

$$\frac{5 + 4x}{x + 3} = \frac{5 - 12}{0}$$

$$= \frac{-7}{0^+} = +\infty$$

$$\lim_{x \rightarrow -3^-}$$

$$\frac{5 + 4x}{x + 3} = \frac{5 - 12}{0}$$

$$= \frac{-7}{0^-} = -\infty$$

$\Rightarrow \therefore x = -3$ is V.A

Find the horizontal and vertical asymptote(s) for

$$f(x) = \frac{|x|}{2x-3}$$

past exam

Horizontal asymptote

$$\lim_{x \rightarrow \infty} \frac{x}{2x-3} \quad \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow \infty} \frac{x}{x(2 - \frac{3}{x})}$$

$$= \frac{1}{2 - \frac{3}{\infty}} = \frac{1}{2-0} = \frac{1}{2}$$

$$\lim_{x \rightarrow -\infty} \frac{-x}{2x-3} \quad \frac{-\infty}{-\infty}$$

$$\lim_{x \rightarrow -\infty} \frac{-x}{x(2 - \frac{3}{x})}$$

$$= \frac{-1}{2 - \frac{3}{-\infty}} = \frac{-1}{2-0} = -\frac{1}{2}$$

$\therefore y = \frac{1}{2} \quad y = -\frac{1}{2}$ are H.A's

Vertical asymptote

$$2x-3=0 \longrightarrow 2x=3 \longrightarrow x=\frac{3}{2}$$

$$\lim_{x \rightarrow \frac{3}{2}^+} \frac{x}{2x-3}$$

$$= \frac{\frac{3}{2}}{0^+} = +\infty$$

$$\lim_{x \rightarrow \frac{3}{2}^-} \frac{x}{2x-3}$$

$$= \frac{\frac{3}{2}}{0^-} = -\infty$$

$\therefore x = \frac{3}{2}$ is V.A

$$\text{Let } f(x) = \frac{\sqrt{x^2 + 5}}{2x + 3}$$

Apr 2017

Find the horizontal and vertical asymptotes of the graph of f .

Horizontal Asymptote

$$\lim_{x \rightarrow \pm\infty} \frac{\sqrt{x^2 + 5}}{2x + 3} \rightarrow \frac{\infty}{\pm\infty}$$

$$\lim_{x \rightarrow \pm\infty} \frac{\sqrt{x^2(1 + \frac{5}{x^2})}}{2x + 3} = \lim_{x \rightarrow \pm\infty} \frac{\sqrt{x^2} \sqrt{1 + \frac{5}{x^2}}}{2x + 3}$$

$$\lim_{x \rightarrow \pm\infty} \frac{|x| \sqrt{1 + \frac{5}{x^2}}}{2x + 3}$$

$$\lim_{x \rightarrow \infty} \frac{x \sqrt{1 + \frac{5}{x^2}}}{2x + 3}$$

$$\lim_{x \rightarrow -\infty} \frac{-x \sqrt{1 + \frac{5}{x^2}}}{2x + 3}$$

$$\lim_{x \rightarrow \infty} \frac{\cancel{x} \sqrt{1 + \frac{5}{x^2}}}{\cancel{x} (2 + \frac{3}{x})}$$

$$\lim_{x \rightarrow -\infty} \frac{\cancel{-x} \sqrt{1 + \frac{5}{x^2}}}{\cancel{-x} (2 + \frac{3}{x})}$$

$$= \frac{\sqrt{1+0}}{2+0} = \frac{1}{2} \quad \quad \quad = \frac{-\sqrt{1+0}}{2+0} = -\frac{1}{2}$$

$\Rightarrow \therefore y = \frac{1}{2}$ and $y = -\frac{1}{2}$ are HA's

Vertical Asymptote

$$\lim_{x \rightarrow -\frac{3}{2}^+} \frac{\sqrt{x^2 + 5}}{2x + 3}$$

$$= \frac{\sqrt{9/4 + 5}}{0^+} = +\infty$$

$$2x + 3 = 0 \rightarrow 2x = -3 \rightarrow x = -\frac{3}{2}$$

$$\lim_{x \rightarrow -\frac{3}{2}^-} \frac{\sqrt{x^2 + 5}}{2x + 3}$$

$$= \frac{\sqrt{9/4 + 5}}{0^-} = -\infty$$

$\Rightarrow \therefore x = -\frac{3}{2}$ is V.A



Text book question



Find the horizontal and vertical asymptotes (if any) of the following function

$$y = \frac{x^3 - x}{x^2 - 6x + 5}$$

Horizontal Asymptote

$$\lim_{x \rightarrow \pm\infty} \frac{x^3 - x}{x^2 - 6x + 5} = \frac{\pm\infty}{\infty}$$

$$\lim_{x \rightarrow \pm\infty} \frac{x^3 \left[1 - \frac{1}{x^2} \right]}{x^2 \left[1 - \frac{6}{x} + \frac{5}{x^2} \right]} = \frac{\pm\infty [1-0]}{1-0+0} = \pm\infty$$

$\Rightarrow \therefore$ No horizontal Asymptote.

Vertical Asymptote

$$x=1$$

$$x^2 - 6x + 5 = 0$$

$$(x-1)(x-5) = 0 \quad \begin{matrix} \nearrow x=1 \\ \searrow x=5 \end{matrix}$$

$$\lim_{x \rightarrow 1+} \frac{x^3 - x}{x^2 - 6x + 5}$$

$$\lim_{x \rightarrow 1+} \frac{x(x^2 - 1)}{(x-1)(x-5)}$$

$$\lim_{x \rightarrow 1+} \frac{x(x-1)(x+1)}{(x-1)(x-5)} = \frac{(1)(2)}{-4} = -\frac{1}{2}$$

$$\lim_{x \rightarrow 1-} \frac{x^3 - x}{x^2 - 6x + 5}$$

$$\lim_{x \rightarrow 1-} \frac{x(x^2 - 1)}{(x-1)(x-5)}$$

$$\lim_{x \rightarrow 1-} \frac{x(x-1)(x+1)}{(x-1)(x-5)} = \frac{2}{-4} = -\frac{1}{2}$$

$\Rightarrow \therefore x=1$ is Not V.A

See Next page p2!

$$\underline{\underline{x = 5}}$$

$$\lim_{x \rightarrow 5^+} \frac{x^3 - x}{x^2 - 6x + 5}$$

$$\lim_{x \rightarrow 5^+} \frac{x(x^2 - 1)}{(x-1)(x-5)}$$

$$\lim_{x \rightarrow 5^+} \frac{x \cancel{(x-1)} (x+1)}{\cancel{(x-1)} (x-5)}$$

$$= \frac{5(6)}{0^+} = +\infty$$

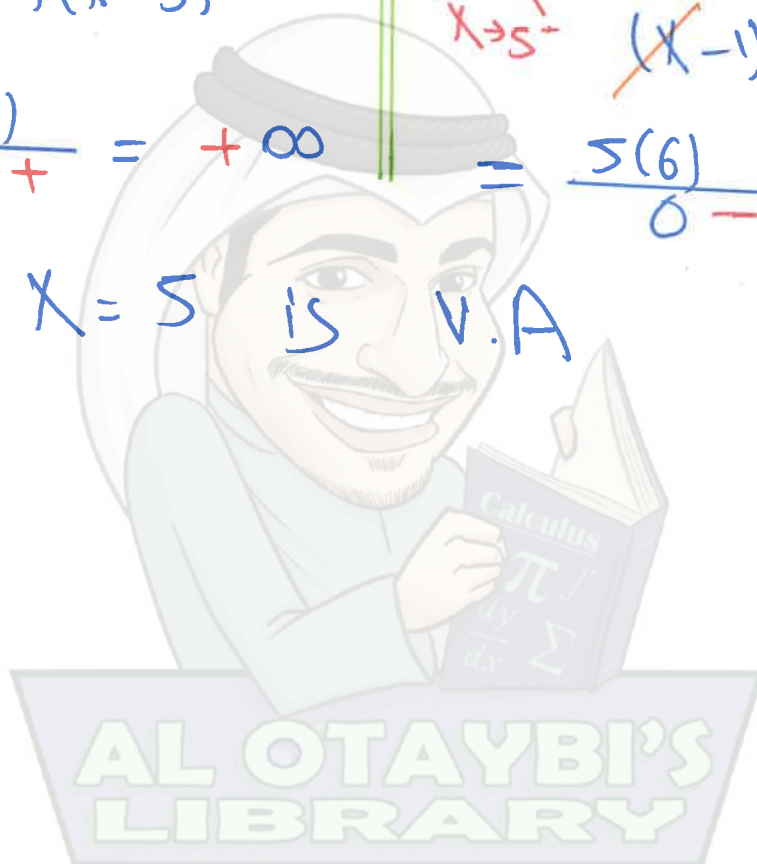
$$\lim_{x \rightarrow 5^-} \frac{x^3 - x}{x^2 - 6x + 5}$$

$$\lim_{x \rightarrow 5^-} \frac{x(x^2 - 1)}{(x-1)(x-5)}$$

$$\lim_{x \rightarrow 5^-} \frac{x \cancel{(x-1)} (x+1)}{\cancel{(x-1)} (x-5)}$$

$$= \frac{5(6)}{0^-} = -\infty$$

$\Rightarrow \therefore x = 5$ is V.A



Find the horizontal and vertical asymptote(s) for

$$f(x) = \frac{2x\sqrt{x^2+1}}{x^2+3x}$$

Horizontal asymptote

$$\lim_{x \rightarrow \pm\infty} \frac{2x\sqrt{x^2+1}}{x^2+3x} \quad \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow \pm\infty} \frac{2x\sqrt{x^2(1+\frac{1}{x^2})}}{x^2+3x}$$

$$\lim_{x \rightarrow \pm\infty} \frac{2x|x|\sqrt{1+\frac{1}{x^2}}}{x^2+3x}$$

$$\lim_{x \rightarrow \infty} \frac{2x^2\sqrt{1+\frac{1}{x^2}}}{x^2+3x}$$

$$\lim_{x \rightarrow -\infty} \frac{-2x^2\sqrt{1+\frac{1}{x^2}}}{x^2+3x}$$

$$\lim_{x \rightarrow \infty} \frac{2x^2\sqrt{1+\frac{1}{x^2}}}{x^2(1+\frac{3}{x})}$$

$$\lim_{x \rightarrow -\infty} \frac{-2x^2\sqrt{1+\frac{1}{x^2}}}{x^2(1+\frac{3}{x})}$$

$$= \frac{2\sqrt{1+0}}{1+0} = 2$$

$$= \frac{-2\sqrt{1+0}}{1+0} = -2$$

* $y = 2$, $y = -2$ are HA's

Vertical asymptote

$$x^2 + 3x = 0$$
$$x(x+3) = 0$$
$$x = 0 \quad x = -3$$

at $x=0$

$$\lim_{x \rightarrow 0^+} \frac{2x \sqrt{x^2+1}}{x^2+3x} = \frac{0}{0}$$

$$\lim_{x \rightarrow 0^+} \frac{2x \sqrt{x^2+1}}{\cancel{x}(x+3)}$$

$$= \frac{2\sqrt{0+1}}{0+3} = \frac{2}{3}$$

$$\lim_{x \rightarrow 0^-} \frac{2x \sqrt{x^2+1}}{x^2+3x}$$

$$\lim_{x \rightarrow 0^-} \frac{2x \sqrt{x^2+1}}{\cancel{x}(x+3)}$$

$$= \frac{2\sqrt{0+1}}{0+3} = \frac{2}{3}$$

$\therefore x=0$ is Not V.A

at $x=-3$

$$\lim_{x \rightarrow -3^+} \frac{2x \sqrt{x^2+1}}{x^2+3x}$$

$$\lim_{x \rightarrow -3^+} \frac{2x \sqrt{x^2+1}}{\cancel{x}(x+3)}$$

$$= \frac{2\sqrt{9+1}}{0^+} = +\infty$$

$$\lim_{x \rightarrow -3^-} \frac{2x \sqrt{x^2+1}}{x^2+3x}$$

$$\lim_{x \rightarrow -3^-} \frac{2x \sqrt{x^2+1}}{\cancel{x}(x+3)}$$

$$= \frac{2\sqrt{9+1}}{0^-} = -\infty$$

$\therefore x=-3$ is V.A

Find the horizontal and vertical asymptote(s) for

$$f(x) = \frac{2x + \sqrt{x^2 + 1}}{x + 2}$$

Horizontal asymptote

$$\lim_{x \rightarrow \pm\infty} \frac{2x + \sqrt{x^2 + 1}}{x + 2}$$

$$\lim_{x \rightarrow \pm\infty} \frac{2x + \sqrt{x^2(1 + \frac{1}{x^2})}}{x + 2}$$

$$\lim_{x \rightarrow \pm\infty} \frac{2x + |x| \sqrt{1 + \frac{1}{x^2}}}{x + 2}$$

$$\lim_{x \rightarrow \infty} \frac{2x + x \sqrt{1 + \frac{1}{x^2}}}{x + 2}$$

$$\lim_{x \rightarrow \infty} \frac{x [2 + \sqrt{1 + \frac{1}{x^2}}]}{x [1 + \frac{2}{x}]}$$

$$= \frac{2 + \sqrt{1+0}}{1+0}$$

$$= \frac{2+1}{1} = \textcircled{3}$$

$$\lim_{x \rightarrow -\infty} \frac{2x - x \sqrt{1 + \frac{1}{x^2}}}{x + 2}$$

$$\lim_{x \rightarrow -\infty} \frac{x [2 - \sqrt{1 + \frac{1}{x^2}}]}{x [1 + \frac{2}{x}]}$$

$$= \frac{2 - \sqrt{1+0}}{1+0}$$

$$= \frac{2-1}{1} = \textcircled{1}$$

∴ $y = 3$ $y = 1$ are HA's

Vertical asymptote

$$\begin{aligned}x+2 &= 0 \\x &= -2\end{aligned}$$

$$\lim_{x \rightarrow -2^+} \frac{2x + \sqrt{x^2 + 1}}{x + 2}$$

$$= \frac{-4 + \sqrt{5}}{0^+}$$

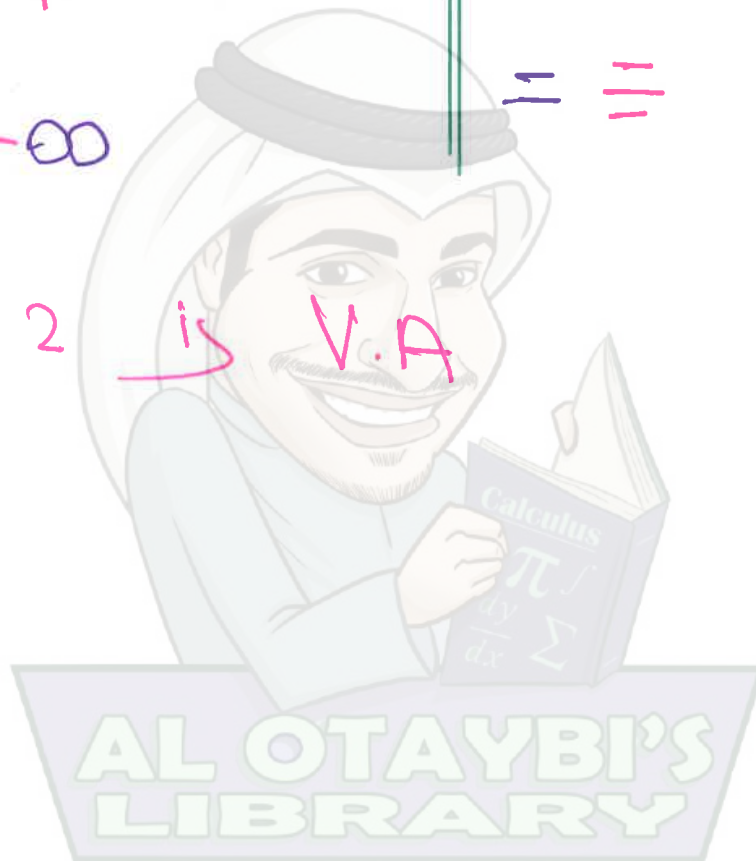
$$= \frac{+}{+} = +\infty$$

$$\lim_{x \rightarrow -2^-} \frac{2x + \sqrt{x^2 + 1}}{x + 2}$$

$$= \frac{-4 + \sqrt{5}}{0^-}$$

$$= \frac{+}{-} = -\infty$$

$\therefore x = -2$ is V.A



Find the horizontal and vertical asymptote(s) for

$$f(x) = \frac{e^{2x}}{2e^{2x} - 2}$$

Horizontal asymptote

$$\lim_{x \rightarrow \infty} \frac{e^{2x}}{2e^{2x} - 2} \quad \frac{e^\infty}{2e^\infty - 2} \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow -\infty} \frac{e^{2x}}{2e^{2x} - 2}$$

$$\lim_{x \rightarrow \infty} \frac{\cancel{e^{2x}}}{\cancel{e^{2x}} \left[2 - \frac{2}{e^{2x}} \right]} = \frac{1}{2 - \frac{2}{\infty}} = \frac{1}{2} = \left(\frac{1}{2} \right)$$

$$= \frac{e^{-\infty}}{2e^{-\infty} - 2} = \frac{0}{0 - 2} = \frac{0}{-2} = 0 = (0)$$

∴ $y = \frac{1}{2}$ and $y = 0$ are H.A.'s

Vertical asymptote

$$2e^{2x} - 2 = 0 \rightarrow 2e^{2x} = 2 \rightarrow e^{2x} = 1 \rightarrow \ln e^{2x} = \ln 1$$

$$2x = 0$$

$$x = 0$$

$$\lim_{x \rightarrow 0^+} \frac{e^{2x}}{2e^{2x} - 2} = \frac{1}{0^+} = +\infty$$

$$\lim_{x \rightarrow 0^-} \frac{e^{2x}}{2e^{2x} - 2} = \frac{1}{0^-} = -\infty$$

Eng. Mohammad Alotaybi 90048657 Twitter: @M_Alotaybi

⇒ ∴ $x = 0$ is V.A

Find the horizontal and vertical asymptote(s) for

$$f(x) = \frac{2e^x}{e^x - 2}$$

Horizontal asymptote

$$\lim_{x \rightarrow \infty} \frac{2e^x}{e^x - 2} \quad \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow \infty} \frac{2e^x}{e^x [1 - \frac{2}{e^x}]}$$

$$= \frac{2}{1 - \frac{2}{\infty}} = \frac{2}{1-0} = 2$$

$$\lim_{x \rightarrow -\infty} \frac{2e^x}{e^x - 2}$$

$$= \frac{2e^{-\infty}}{e^{-\infty} - 2}$$

$$= \frac{0}{0-2} = 0$$

$\Rightarrow \therefore y=2$ $y=0$ are HA's

Vertical asymptote

$$e^x - 2 = 0 \rightarrow e^x = 2 \rightarrow \ln e^x = \ln 2 \rightarrow x = \ln 2$$

$$\lim_{x \rightarrow (\ln 2)^+} \frac{2e^x}{e^x - 2}$$

$$= \frac{2e^{\ln 2}}{0^+} = +\infty$$

$$\lim_{x \rightarrow (\ln 2)^-} \frac{2e^x}{e^x - 2}$$

$$= \frac{2e^{\ln 2}}{0^-} = -\infty$$

$\Rightarrow \therefore x = \ln 2$ is V.A.

Find the horizontal and vertical asymptote(s) for

$$f(x) = \frac{-3x^2 - 2}{2x^2 + x + 5}$$

Horizontal asymptote

$$\lim_{x \rightarrow \pm\infty} \frac{-3x^2 - 2}{2x^2 + x + 5} \quad \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow \pm\infty} \frac{\cancel{x^2} [-3 - \frac{2}{x^2}]}{\cancel{x^2} [2 + \frac{1}{x} + \frac{5}{x^2}]} = \frac{-3 - 0}{2 + 0 + 0} = -\frac{3}{2}$$

$\Rightarrow \therefore y = -\frac{3}{2}$ is H.A

Vertical asymptote

بطل المقام

$$2x^2 + x + 5 = 0$$

و ناقصا

$$a=2 \quad b=1 \quad c=5$$

دعمين

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-1 \pm \sqrt{(1)^2 - 4(2)(5)}}{2(2)} = \frac{-1 \pm \sqrt{1 - 40}}{4}$$

$$= \frac{-1 \pm \sqrt{-39}}{-4}$$

impossible

$\therefore$ No V.A.

$$\text{Let } f(x) = \frac{x^2 - 4x}{|x|(x+2)}$$

July 2017

(a) Find and classify the discontinuities of f as removable, infinite or jump.

(b) Find the horizontal and vertical asymptotes of the graph of f , if any.

$$|x|(x+2) = 0 \quad \begin{matrix} \rightarrow x=0 \\ \rightarrow x=-2 \end{matrix}$$

at $x=0$

$f(0)$ undefined

$$\lim_{x \rightarrow 0} \frac{x^2 - 4x}{|x|(x+2)}$$

$$\lim_{x \rightarrow 0^+} \frac{\cancel{x}(x-4)}{+\cancel{x}(x+2)}$$

$$= \frac{-4}{2} = -2$$

$$\lim_{x \rightarrow 0^-} \frac{\cancel{x}(x-4)}{-\cancel{x}(x+2)}$$

$$= \frac{-4}{-2} = 2$$

$\therefore$ at $x=0$ $\rightarrow$ jump discontinuity

at $x=-2$

$f(-2)$ undefined

$$\lim_{x \rightarrow -2} \frac{x^2 - 4x}{|x|(x+2)}$$

$$= \lim_{x \rightarrow -2} \frac{\cancel{x}(x-4)}{-\cancel{x}(x+2)}$$

$$\lim_{x \rightarrow -2^+} \frac{x-4}{-(x+2)} = \frac{-6}{0^-} = +\infty$$

$$\lim_{x \rightarrow -2^-} \frac{x-4}{-(x+2)} = \frac{-6}{0^+} = -\infty$$

$\Rightarrow$ at $x=-2$ $\rightarrow$ infinite discontinuity.

b) Horizontal Asymptotes

$$\lim_{x \rightarrow \pm\infty} \frac{x^2 - 4x}{x(x+2)}$$

$$\lim_{x \rightarrow \infty} \frac{x^2 - 4x}{x(x+2)}$$

$$\lim_{x \rightarrow \infty} \frac{\cancel{x^2} \left[1 - \frac{4}{x} \right]}{\cancel{x} \cdot x \left(1 + \frac{2}{x} \right)}$$

$$= \frac{1-0}{1+0} = 1$$

$$\lim_{x \rightarrow -\infty} \frac{x^2 - 4x}{-x(x+2)}$$

$$\lim_{x \rightarrow -\infty} \frac{\cancel{x^2} \left[1 - \frac{4}{x} \right]}{-\cancel{x} x \left(1 + \frac{2}{x} \right)}$$

$$= \frac{1-0}{-(1+0)} = -1$$

$\Rightarrow \therefore y=1$ and $y=-1$ are HA's

Vertical Asymptote

From part (a)
 $x = -2$ is V.A



Homework



Find the horizontal and vertical asymptote(s) for

$$f(x) = \frac{x|x|}{x^2 - x}$$

H.A $\rightarrow y = 1$ and $y = -1$, V.A $\rightarrow x = 1$

$$\text{Let } f(x) = \begin{cases} e^x - \frac{3}{x-7} + 5, & \text{if } x \leq 1 \\ \ln(x-1) + 3, & \text{if } 1 < x \leq 3 \\ \frac{x^2+1}{x^3+1}, & \text{if } x > 3 \end{cases}$$

Mar 2019

(a) Find $\lim_{x \rightarrow 1} f(x)$, if it exists

$$\lim_{x \rightarrow 1^+} [\ln(x-1) + 3] = \ln 0 + 3 = -\infty$$

$$\lim_{x \rightarrow 1^-} [e^x - \frac{3}{x-7} + 5] = e^1 - \frac{3}{-6} + 5 = e + \frac{1}{2} + 5$$

$$\therefore \lim_{x \rightarrow 1^+} f(x) \neq \lim_{x \rightarrow 1^-} f(x) \rightarrow \therefore \lim_{x \rightarrow 1} f(x) \text{ DNE}$$

(b) Find $\lim_{x \rightarrow 3} f(x)$, if it exists

$$\lim_{x \rightarrow 3^+} \frac{x^2+1}{x^3+1} = \frac{9+1}{27+1} = \frac{10}{28} = \frac{5}{14}$$

$$\lim_{x \rightarrow 3^-} [\ln(x-1) + 3] = \ln 2 + 3$$

$$\therefore \lim_{x \rightarrow 3^+} f(x) \neq \lim_{x \rightarrow 3^-} f(x) \rightarrow \therefore \lim_{x \rightarrow 3} f(x) \text{ DNE}$$

(c) Find $\lim_{x \rightarrow -\infty} f(x)$, if it exists

$$\begin{aligned} \lim_{x \rightarrow -\infty} f(x) &= \lim_{x \rightarrow -\infty} [e^x - \frac{3}{x-7} + 5] \\ &= 0 - 0 + 5 = 5 \end{aligned}$$

(d) Find $\lim_{x \rightarrow \infty} f(x)$, if it exists

$$\begin{aligned} \lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} \frac{x^2+1}{x^3+1} = \lim_{x \rightarrow \infty} \frac{x^2[1+\frac{1}{x^2}]}{x^3[1+\frac{1}{x^3}]} \\ &= \frac{[1+0]}{\infty[1+0]} = 0 \end{aligned}$$

(e) Find, if any, all vertical asymptotes of the graph of f

Vertical Asymptote

$$x=1 \quad \left(\lim_{x \rightarrow 1^+} f(x) = -\infty \quad \lim_{x \rightarrow 1^-} f(x) = e + \frac{1}{2} + 5 \right)$$

(f) Find and classify, if any, all the points of discontinuity

$$x=1 \rightarrow \text{infinite discontinuity} \quad \left(\lim_{x \rightarrow 1^+} f(x) = -\infty, \lim_{x \rightarrow 1^-} f(x) = e + \frac{1}{2} + 5 \right)$$

$$x=3 \rightarrow \text{Jump discontinuity} \quad \left(\lim_{x \rightarrow 3^+} f(x) = \frac{5}{14} \neq \lim_{x \rightarrow 3^-} f(x) = \ln 2 + 3 \right)$$

(g) Find, if any, all vertical asymptotes of the graph of f

$$y=5 \quad \left(\lim_{x \rightarrow -\infty} f(x) = 5 \right)$$

$$y=0 \quad \left(\lim_{x \rightarrow \infty} f(x) = 0 \right)$$

Use the given graph of f to evaluate each of the following limits, if it exists.

Mar 2018

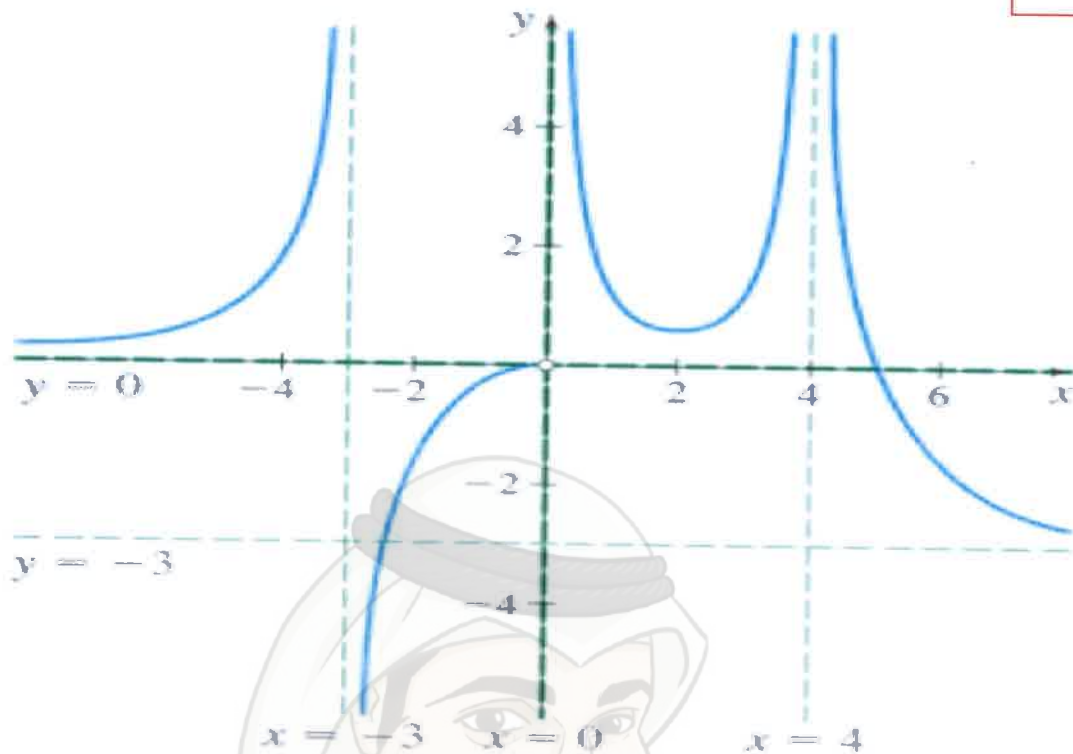


Figure 1: The graph of $y = f(x)$.

(a) $\lim_{x \rightarrow -3} f(x)$

$\lim_{x \rightarrow -3^+} f(x) = -\infty$

$\lim_{x \rightarrow -3^-} f(x) = \infty$

$\therefore \lim_{x \rightarrow -3^+} f(x) \neq \lim_{x \rightarrow -3^-} f(x)$

$\therefore \lim_{x \rightarrow -3} f(x) \text{ DNE}$

(b) $\lim_{x \rightarrow 0^-} f(x)$

$= 0$

(c) $\lim_{x \rightarrow 0^+} f(x)$

$= \infty$

(d) $\lim_{x \rightarrow -\infty} f(x)$

$= 0$

يفرق الابطالين: -3 0 4
يفرق الصورة:



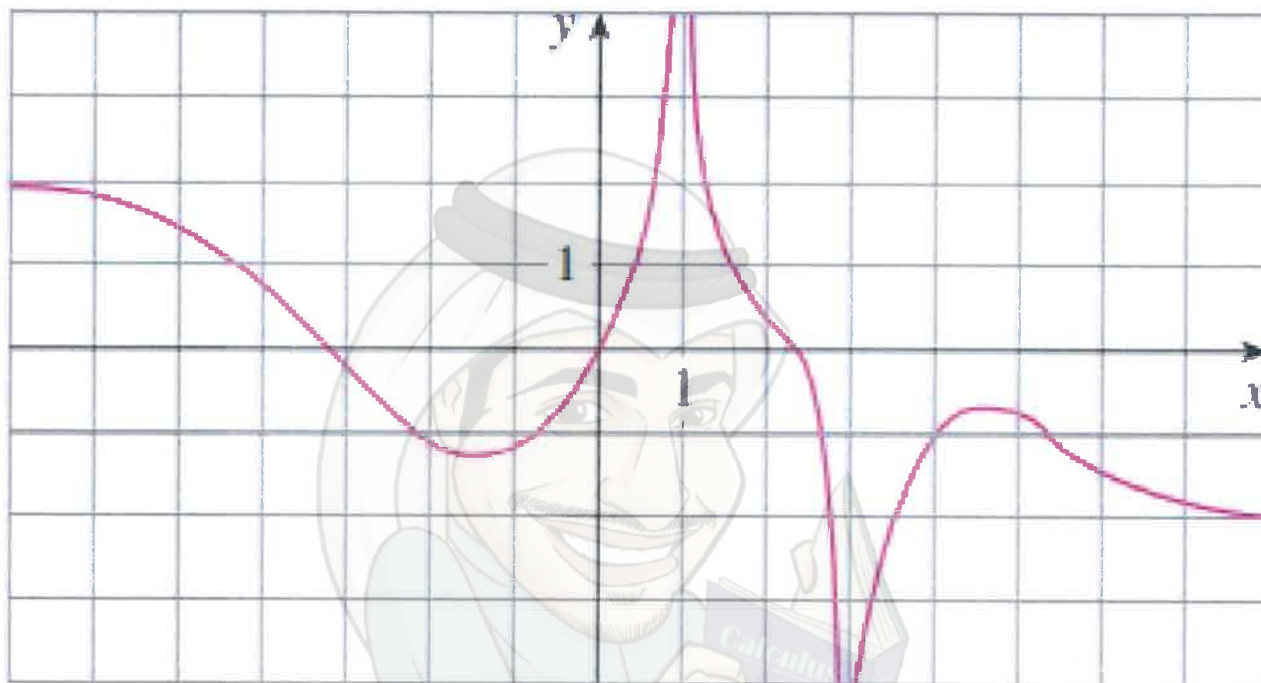
Text book question



For the function f whose graph is given, state the following

(a) $\lim_{x \rightarrow \infty} f(x)$ (b) $\lim_{x \rightarrow -\infty} f(x)$ (c) $\lim_{x \rightarrow 1} f(x)$ (d) $\lim_{x \rightarrow 3} f(x)$

(e) The equation of the asymptotes.



a) $\lim_{x \rightarrow \infty} f(x) = -2$

يفرق الدتجاهين : 1, 3
يفرق الصورق :

b) $\lim_{x \rightarrow -\infty} f(x) = -2$

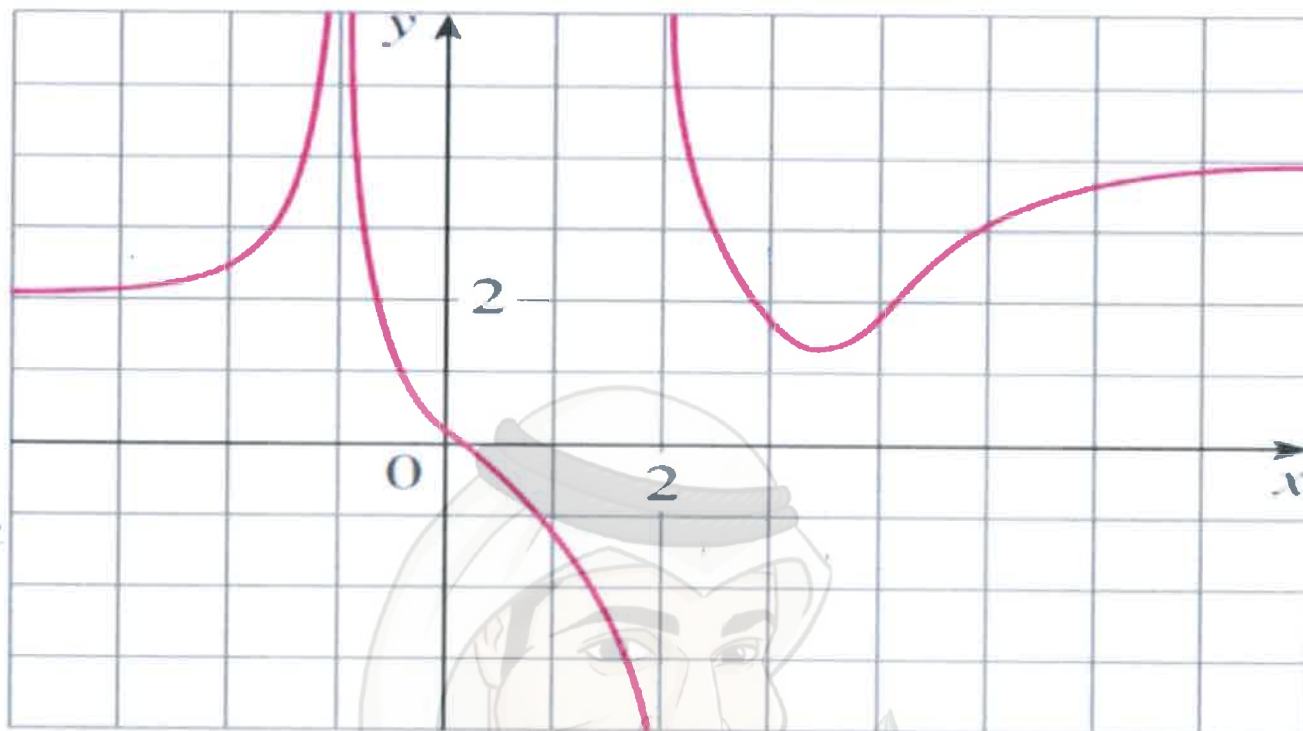
c) $\lim_{x \rightarrow 1} f(x) = +\infty$ ($\lim_{x \rightarrow 1^+} f(x) = \infty$, $\lim_{x \rightarrow 1^-} f(x) = \infty$)

d) $\lim_{x \rightarrow 3} f(x) = -\infty$ ($\lim_{x \rightarrow 3^+} f(x) = -\infty$, $\lim_{x \rightarrow 3^-} f(x) = -\infty$)

e) Equation of horizontal Asymptote $\rightarrow$ $y = -2$ ($\lim_{x \rightarrow \infty} f(x) = -2$)
 $y = 2$ ($\lim_{x \rightarrow -\infty} f(x) = 2$)

Equation of Vertical Asymptote $\rightarrow$ $x = 1$ ($\lim_{x \rightarrow 1^+} f(x) = \infty$, $\lim_{x \rightarrow 1^-} f(x) = \infty$)
 $x = 3$ ($\lim_{x \rightarrow 3^+} f(x) = -\infty$, $\lim_{x \rightarrow 3^-} f(x) = -\infty$)

Using the given graph of the function f shown below to find all vertical and horizontal asymptotes to the graph of f



Oct 2018

$$\lim_{x \rightarrow -1^+} f(x) = \infty$$

يُفوق الاتجاهين : -1, 2
يُفوق الممرّة :

$$\lim_{x \rightarrow -1^-} f(x) = \infty$$

$\therefore x = -1$ is Vertical Asymptote

$$\lim_{x \rightarrow 2^+} f(x) = \infty$$

$$\lim_{x \rightarrow 2^-} f(x) = -\infty$$

$\therefore x = 2$ is Vertical Asymptote

Horizontal Asymptotes

$$\lim_{x \rightarrow \infty} f(x) = 4$$

$$\lim_{x \rightarrow -\infty} f(x) = 2$$

$y = 4, y = 2$ are H.A's

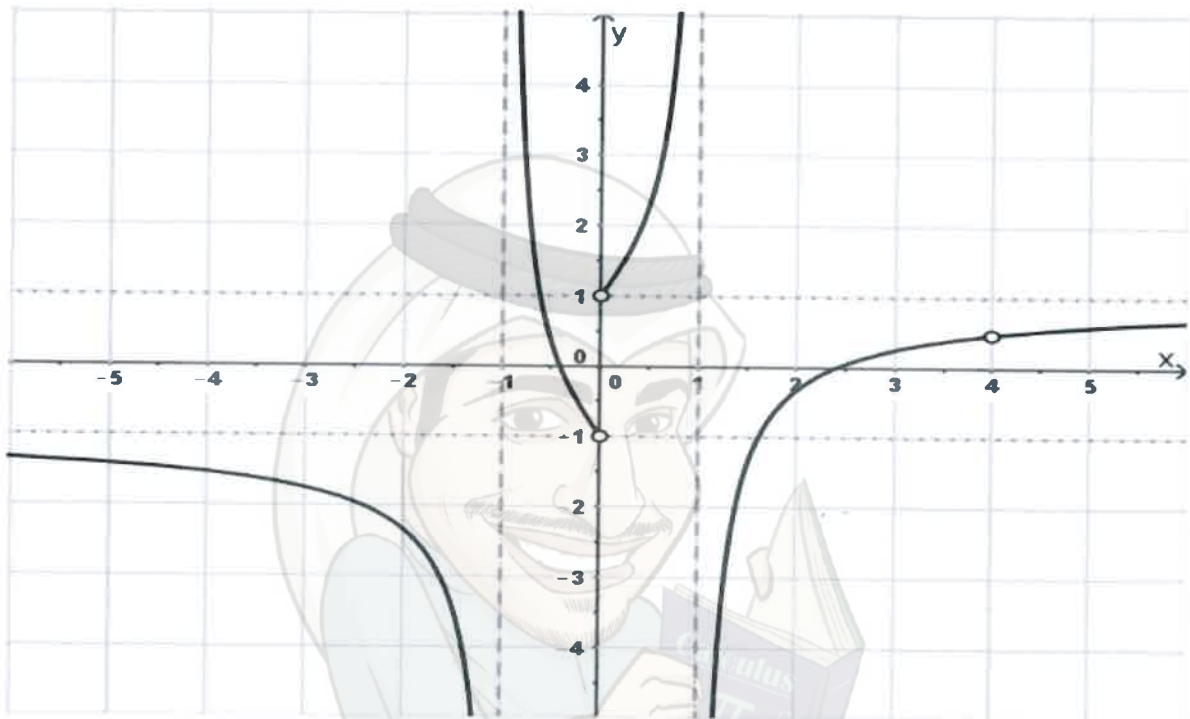
Use the graph of $y = f(x)$ shown below to :

April 2017

a) Find the following limits:

i) $\lim_{x \rightarrow -1^+} f(x)$ ii) $\lim_{x \rightarrow 0^+} f(x)$ iii) $\lim_{x \rightarrow 0^-} f(x)$ iv) $\lim_{x \rightarrow 1^+} f(x)$ v) $\lim_{x \rightarrow \infty} f(x)$

b) Find and classify the discontinuities of f as removable, infinite or jump.



i) $\lim_{x \rightarrow -1^+} f(x) = \infty$ ii) $\lim_{x \rightarrow 0^+} f(x) = 1$ iii) $\lim_{x \rightarrow 0^-} f(x) = -1$

iv) $\lim_{x \rightarrow 1^+} f(x) = -\infty$ v) $\lim_{x \rightarrow \infty} f(x) = 1$

نقطة الانكسار:

1, 0, -1

نقطة الصورة:

4

b)

at $x=4 \rightarrow$ removable disc. ($f(4) \neq \lim_{x \rightarrow 4} f(x)$)

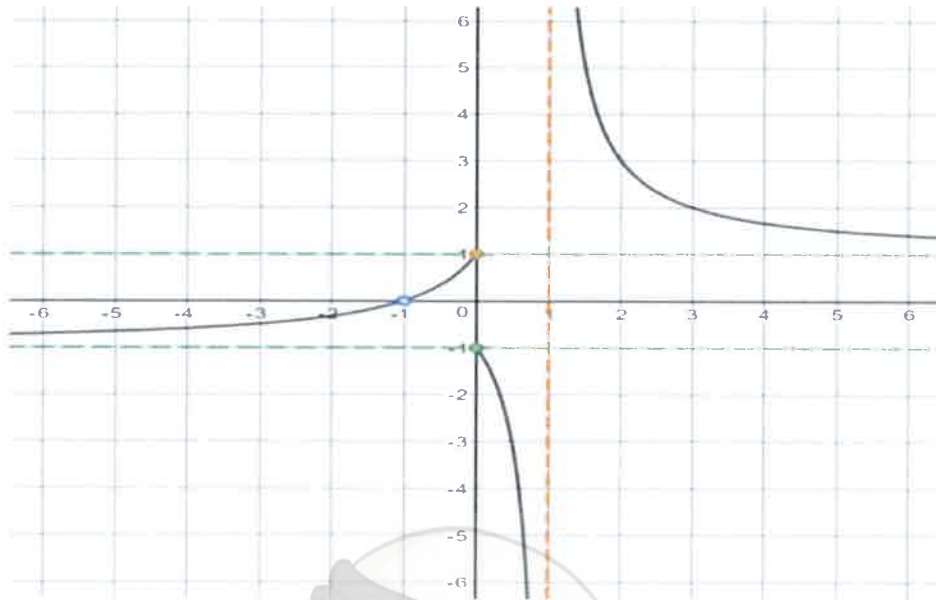
at $x=0 \rightarrow$ Jump disc. ($\lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$)
 $1 \neq -1$

at $x=1 \rightarrow$ infinite disc. ($\lim_{x \rightarrow 1^+} f(x) = -\infty$, $\lim_{x \rightarrow 1^-} f(x) = \infty$)

at $x=-1 \rightarrow$ infinite disc. ($\lim_{x \rightarrow -1^+} f(x) = \infty$, $\lim_{x \rightarrow -1^-} f(x) = -\infty$)

Use the given graph of f to select the correct answer

July 2017



يفرق الدباجين

1, 0

يفرق المبررة

-1

I) $\lim_{x \rightarrow 0} f(x) =$

- a) 1 b) 0 c) ∞ d) -1 e) None of the above

$\lim_{x \rightarrow 0^+} f(x) = -1$

$\lim_{x \rightarrow 0^-} f(x) = 1$

$\therefore \lim_{x \rightarrow 0} f(x) \text{ DNE}$

(e) ✓

II) $\lim_{x \rightarrow 1^+} f(x) =$

- a) $+\infty$ b) $-\infty$ c) 1 d) 0 e) None of the above

∞ (a) ✓

III) The graph of f has

- a) no vertical asymptote b) only one vertical asymptote
c) only one horizontal asymptote d) no horizontal asymptote
e) None of the above

$x=1$ is V.A

$\lim_{x \rightarrow \infty} f(x) = -1$

$\lim_{x \rightarrow -\infty} f(x) = 1$

$y=-1$
 $y=1$ are H.A's

(b) ✓

III) The function f has

- a) a jump discontinuity at $x=1$ b) a removable discontinuity at $x=0$
c) a jump discontinuity at $x=1$ d) a removable discontinuity at $x=-1$
e) None of the above

at $x=-1 \rightarrow$ removable disc.

at $x=0 \rightarrow$ Jump disc.

at $x=1 \rightarrow$ infinite disc.

(d) ✓

2.7 – 2.8 Derivatives by definition

المشتقة بالتعريف

General derivative

(المشتقة العامة $f'(x)$)

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Derivative at a point a

(المشتقة عند نقطة $f'(a)$)

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

Or

$$f'(a) = \lim_{h \rightarrow a} \frac{f(a+h) - f(a)}{h}$$

Easy Questions

State the definition of the general derivative of the function f

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

State the definition of the derivative of the function f at $x = a$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

Use definition of derivative to find $f'(x)$

if $f(x) = \sqrt{x+5}$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{x+h+5} - \sqrt{x+5}}{h} \quad \frac{0}{0} \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{x+h+5} - \sqrt{x+5}}{h} * \frac{\sqrt{x+h+5} + \sqrt{x+5}}{\sqrt{x+h+5} + \sqrt{x+5}} \\ &= \lim_{h \rightarrow 0} \frac{x+h+5 - (x+5)}{h (\sqrt{x+h+5} + \sqrt{x+5})} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{x} + h + \cancel{5} - \cancel{x} - \cancel{5}}{h (\sqrt{x+h+5} + \sqrt{x+5})} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{h}}{\cancel{h} (\sqrt{x+h+5} + \sqrt{x+5})} \\ &= \frac{1}{\sqrt{x+0+5} + \sqrt{x+5}} = \frac{1}{2\sqrt{x+5}} \end{aligned}$$

Use the definition of derivative to find $f'(x)$ of

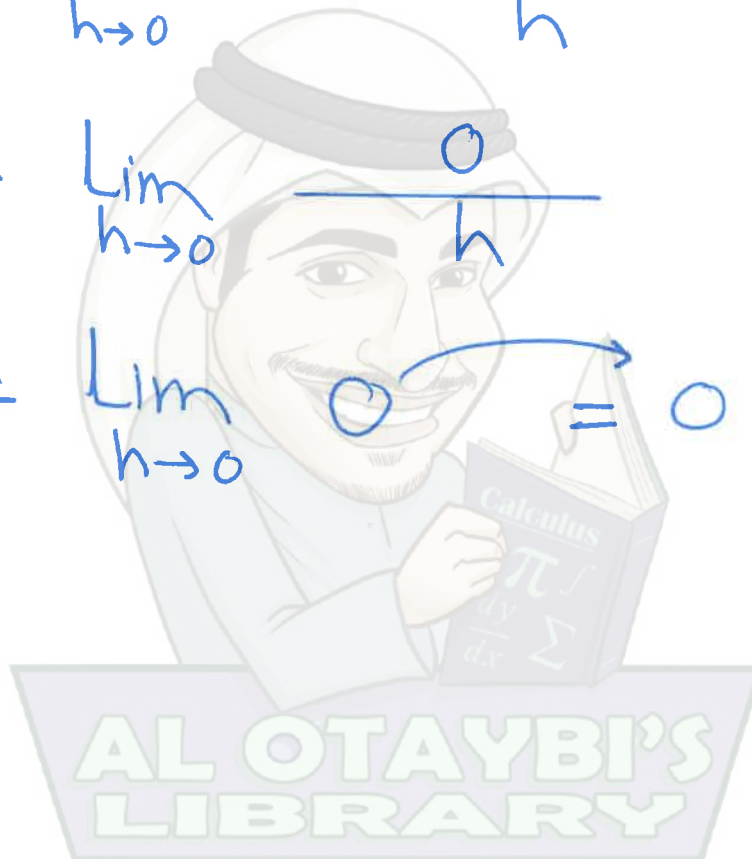
$$f(x) = \pi$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\pi - \pi}{h}$$

$$= \lim_{h \rightarrow 0} \frac{0}{h}$$

$$= \lim_{h \rightarrow 0} 0 = 0$$



Use the definition of derivative to find $f'(x)$ of

$$f(x) = \frac{1}{\sqrt{x}}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{\sqrt{x+h}} - \frac{1}{\sqrt{x}}}{h} \quad \frac{\frac{1}{\sqrt{x+0}} - \frac{1}{\sqrt{x}}}{0} \quad \frac{0}{0}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{\sqrt{x}}{\sqrt{x+h} \sqrt{x}} - \frac{\sqrt{x+h}}{\sqrt{x} \sqrt{x+h}}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{\sqrt{x} - \sqrt{x+h}}{\sqrt{x+h} \sqrt{x}}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{x} - \sqrt{x+h}}{h \sqrt{x+h} \sqrt{x}} * \frac{\sqrt{x} + \sqrt{x+h}}{\sqrt{x} + \sqrt{x+h}}$$

$$= \lim_{h \rightarrow 0} \frac{x - [x+h]}{h \sqrt{x+h} \sqrt{x} (\sqrt{x} + \sqrt{x+h})}$$

$$= \lim_{h \rightarrow 0} \frac{x - x - h}{h \sqrt{x+h} \sqrt{x} (\sqrt{x} + \sqrt{x+h})}$$

$$= \lim_{h \rightarrow 0} \frac{-h}{h \sqrt{x+h} \sqrt{x} (\sqrt{x} + \sqrt{x+h})}$$

$$= \frac{-1}{\sqrt{x+0} \sqrt{x} (\sqrt{x} + \sqrt{x+0})} = \frac{-1}{\sqrt{x} \sqrt{x} (\sqrt{x} + \sqrt{x})} = \frac{-1}{x \cdot 2\sqrt{x}}$$

$$\frac{-1}{2x^{\frac{3}{2}}}$$

Use the definition of derivative to find the derivative of

$$f(x) = x^2 + \sqrt{1-x}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^2 + \sqrt{1-(x+h)} - [x^2 + \sqrt{1-x}]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^2 + \sqrt{1-x-h} - x^2 - \sqrt{1-x}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2 + \sqrt{1-x-h} - \sqrt{1-x}}{h}$$

$$= \lim_{h \rightarrow 0} \left[\frac{(x+h)^2 - x^2}{h} + \frac{\sqrt{1-x-h} - \sqrt{1-x}}{h} \right]$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \cdot \frac{0}{0} + \lim_{h \rightarrow 0} \frac{\sqrt{1-x-h} - \sqrt{1-x}}{h} \cdot \frac{0}{0}$$

$$= \lim_{h \rightarrow 0} \frac{[(x+h)-x][(x+h)+x]}{h} + \lim_{h \rightarrow 0} \frac{1-x-h - [1-x]}{h(\sqrt{1-x-h} + \sqrt{1-x})}$$

$$= \lim_{h \rightarrow 0} \frac{h[2x+h]}{h}$$

$$= 2x + 0$$

$$+ \lim_{h \rightarrow 0} \frac{1-x-h-1+x}{h(\sqrt{1-x-h} + \sqrt{1-x})}$$

$$+ \lim_{h \rightarrow 0} \frac{-h}{h(\sqrt{1-x-h} + \sqrt{1-x})}$$

$$= \frac{-1}{\sqrt{1-x-0} + \sqrt{1-x}} = \frac{-1}{2\sqrt{1-x}}$$

$$= 2x + \frac{-1}{2\sqrt{1-x}}$$



Text book question



Find the derivative of the function using the definition of derivative. State the domain of the function and the domain of its derivative.

$$f(x) = x + \sqrt{x}$$

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{x+h + \sqrt{x+h} - [x + \sqrt{x}]}{h} \\
 &= \lim_{h \rightarrow 0} \frac{x+h + \sqrt{x+h} - x - \sqrt{x}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\cancel{x} + h - \cancel{x} + \sqrt{x+h} - \sqrt{x}}{h} \\
 &= \lim_{h \rightarrow 0} \left[\frac{h}{h} + \frac{\sqrt{x+h} - \sqrt{x}}{h} \right] \\
 &= \lim_{h \rightarrow 0} \left[1 + \frac{\sqrt{x+h} - \sqrt{x}}{h} \right] \\
 &= \lim_{h \rightarrow 0} 1 + \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} \\
 &= 1 + \lim_{h \rightarrow 0} \frac{\cancel{x+h} - \cancel{x}}{h [\sqrt{x+h} + \sqrt{x}]} \\
 &= 1 + \lim_{h \rightarrow 0} \frac{h}{h [\sqrt{x+h} + \sqrt{x}]} \\
 &= 1 + \frac{1}{\sqrt{x} + \sqrt{x}} = \boxed{1 + \frac{1}{2\sqrt{x}}}
 \end{aligned}$$

Domain of f :
 $x \geq 0$
 $D_f = [0, \infty)$

Domain of f' :
 $x \geq 0$
 $[0, \infty) \cap \mathbb{R} \setminus \{0\}$

 $D_{f'} = (0, \infty)$

Use the definition of derivative to find $f'(2)$

where $f(x) = \sqrt{x^3 + 1}$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$f'(2) = \lim_{x \rightarrow 2} \frac{\sqrt{x^3 + 1} - f(2)}{x - 2}$$

$f(2) = \sqrt{8+1}$
 $= \sqrt{9}$
 $= 3$

$$= \lim_{x \rightarrow 2} \frac{\sqrt{x^3 + 1} - 3}{x - 2} \times \frac{\sqrt{x^3 + 1} + 3}{\sqrt{x^3 + 1} + 3}$$

$$= \lim_{x \rightarrow 2} \frac{x^3 + 1 - 9}{(x - 2)(\sqrt{x^3 + 1} + 3)}$$

$$= \lim_{x \rightarrow 2} \frac{x^3 - 8}{(x - 2)(\sqrt{x^3 + 1} + 3)}$$

$$= \lim_{x \rightarrow 2} \frac{(x - 2)(x^2 + 2x + 4)}{(x - 2)(\sqrt{x^3 + 1} + 3)}$$

$$= \frac{4 + 4 + 4}{\sqrt{9} + 3} = \frac{12}{3 + 3} = \frac{12}{6} = 2$$

Use the definition of derivative to find the derivative of
 $f(x) = \sqrt{5x-2}$ at $x = 1$

$$\begin{aligned} f'(a) &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \\ &= \lim_{x \rightarrow 1} \frac{\sqrt{5x-2} - f(1)}{x-1} \quad f(1) = \sqrt{5-2} = \sqrt{3} \\ &= \lim_{x \rightarrow 1} \frac{\sqrt{5x-2} - \sqrt{3}}{x-1} \quad \frac{0}{0} \quad \times \quad \frac{\sqrt{5x-2} + \sqrt{3}}{\sqrt{5x-2} + \sqrt{3}} \\ &= \lim_{x \rightarrow 1} \frac{5x-2-3}{(x-1)(\sqrt{5x-2} + \sqrt{3})} \\ &= \lim_{x \rightarrow 1} \frac{5x-5}{(x-1)(\sqrt{5x-2} + \sqrt{3})} \\ &= \lim_{x \rightarrow 1} \frac{5(x-1)}{(x-1)(\sqrt{5x-2} + \sqrt{3})} \\ &= \frac{5}{\sqrt{3} + \sqrt{3}} = \frac{5}{2\sqrt{3}} \end{aligned}$$



Text book questions



Find $f'(a)$ using definition

$$f(x) = \sqrt{1-2x}$$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$= \lim_{x \rightarrow a} \frac{\sqrt{1-2x} - \sqrt{1-2a}}{x - a} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow a} \frac{\sqrt{1-2x} - \sqrt{1-2a}}{x - a} * \frac{\sqrt{1-2x} + \sqrt{1-2a}}{\sqrt{1-2x} + \sqrt{1-2a}}$$

$$= \lim_{x \rightarrow a} \frac{1-2x - (1-2a)}{(x-a)(\sqrt{1-2x} + \sqrt{1-2a})}$$

$$= \lim_{x \rightarrow a} \frac{\cancel{1} - 2x - \cancel{1} + 2a}{(x-a)(\sqrt{1-2x} + \sqrt{1-2a})}$$

$$= \lim_{x \rightarrow a} \frac{-2(\cancel{x} - a)}{(\cancel{x} - a)(\sqrt{1-2x} + \sqrt{1-2a})}$$

$$= \frac{-2}{\sqrt{1-2a} + \sqrt{1-2a}} = \frac{-2}{2\sqrt{1-2a}}$$

Use the definition of derivative to find the derivative of
 $f(x) = x + \sqrt{x} - 1$ at $a = 1$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$= \lim_{x \rightarrow 1} \frac{x + \sqrt{x} - 1 - [1 + 1 - 1]}{x - 1}$$

$$= \lim_{x \rightarrow 1} \frac{x + \sqrt{x} - 1 - 1 + 1 + 1 - 1}{x - 1}$$

$$= \lim_{x \rightarrow 1} \frac{x - 1 + \sqrt{x} - 1}{x - 1}$$

$$= \lim_{x \rightarrow 1} \left[\frac{x-1}{x-1} + \frac{\sqrt{x}-1}{x-1} \right]$$

$$= \lim_{x \rightarrow 1} \frac{x-1}{x-1} + \lim_{x \rightarrow 1} \frac{\sqrt{x}-1}{x-1} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow 1} 1 + \lim_{x \rightarrow 1} \frac{\sqrt{x}-1}{x-1} \times \frac{\sqrt{x}+1}{\sqrt{x}+1}$$

$$= 1 + \lim_{x \rightarrow 1} \frac{\cancel{x-1}}{(\cancel{x-1})(\sqrt{x}+1)}$$

$$\frac{1}{\sqrt{1}+1} = \frac{1}{1+1} = \frac{1}{2}$$



$$f'(a) = 1 + \frac{1}{2} \\ = \frac{2}{2} + \frac{1}{2} = \frac{3}{2}$$

Write the definition of the derivative of $f'(x)$ at 1

Then use it to find $f'(1)$ if it exist where

$$f(x) = \sqrt{x^2 - 2x + 1} \quad \text{Justify your answer}$$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$f'(1) = \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1}$$

$$f'(1) = \lim_{x \rightarrow 1} \frac{\sqrt{x^2 - 2x + 1} - f(1)}{x - 1}$$

$f(1) = \sqrt{1 - 2 + 1} = 0$

$$= \lim_{x \rightarrow 1} \frac{\sqrt{x^2 - 2x + 1}}{x - 1} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow 1} \frac{\sqrt{(x-1)(x-1)}}{x-1} = \lim_{x \rightarrow 1} \frac{\sqrt{(x-1)^2}}{x-1}$$

$$= \lim_{x \rightarrow 1} \frac{|x-1|}{x-1}$$

$$\lim_{x \rightarrow 1^+} \frac{(x-1)}{(x-1)}$$

$$\lim_{x \rightarrow 1^+} 1 = 1$$

$$\lim_{x \rightarrow 1^-} \frac{-(x-1)}{x-1}$$

$$\lim_{x \rightarrow 1^-} -1 = -1$$

$$\therefore \lim_{x \rightarrow 1^+} \frac{|x-1|}{x-1} \neq \lim_{x \rightarrow 1^-} \frac{|x-1|}{x-1} \Rightarrow \therefore \lim_{x \rightarrow 1} \frac{|x-1|}{x-1} \text{ DNE}$$

$\Rightarrow \therefore f'(1)$ Does Not exist



Text book questions



Each limit represents the derivatives of some function f at some number a . State such an f and a in each case.

$$1) \lim_{h \rightarrow 0} \frac{e^{-2+h} - e^{-2}}{h}$$

$$f = e^{-2+h}$$

$$a = 0$$

$$2) \lim_{h \rightarrow 0} \frac{\cos(\pi + h) + 1}{h}$$

$$f = \cos(\pi + h)$$

$$a = 0$$

$$3) \lim_{\theta \rightarrow \pi/6} \frac{\sin \theta - \frac{1}{2}}{\theta - \pi/6}$$

$$f = \sin \theta$$

$$a = \frac{\pi}{6}$$

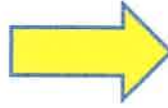


$$f'$$



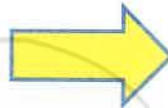
tangent line – normal line – slope
increasing – decreasing – critical numbers

Slope of the **tangent** line



$$m_t = f'(a)$$

Slope of the **normal** line



$$m_n = \frac{1}{f'(a)}$$

tangent line has slope a



$$f'(x) = a$$

Equation of the **tangent** line



$$y - y_o = m_t(x - x_o)$$

Equation of the **normal** line



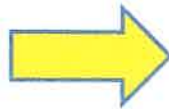
$$y - y_o = m_n(x - x_o)$$

Tangent line is **horizontal**



$$f'(x) = 0$$

Tangent line is **vertical**



$$f'(x) \text{ undefind}$$



الممنوعات



(1) ممنوع أس سالب
(2) ممنوع مقام غير موحد

1) f is continuous at $x = a$

$$2) \lim_{x \rightarrow a} |f'(x)| = \infty$$



Text book questions



Find an equation of the tangent line to the curve at the given point

(Use definition)

$$y = \sqrt{x} \quad , \quad (1, 1)$$

$$\begin{aligned} f'(a) &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \\ &= \lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x - 1} \quad * \quad \frac{\sqrt{x} + 1}{\sqrt{x} + 1} \\ &= \lim_{x \rightarrow 1} \frac{\cancel{x-1}}{(\cancel{x-1})(\sqrt{x} + 1)} = \frac{1}{1+1} = \frac{1}{2} \end{aligned}$$

$\Rightarrow \therefore$ The equation of the tangent line : $y - y_0 = m(x - x_0)$
 $y - 1 = \frac{1}{2}(x - 1)$



Extra



(b) Find the slope of tangent line

(c) Find the slope of the normal line

(d) Find the equation of normal line

(b) The slope of the tangent line = $\frac{1}{2}$

(c) The slope of the normal line = -2

(d) Equation of the normal line :

$$y - y_0 = m_n(x - x_0)$$

$$y - 1 = -2(x - 1)$$



Text book questions



(a) Find the slope of the tangent line to the curve

$y = x - x^2$ at the point $(1, 0)$ using definition.

(b) Find the equation of the tangent line.

$$(a) f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$f'(1) = \lim_{x \rightarrow 1} \frac{(x - x^2) - (1 - 1)}{x - 1}$$

$$= \lim_{x \rightarrow 1} \frac{x - x^2 - 1 + 1}{x - 1}$$

$$= \lim_{x \rightarrow 1} \frac{x - 1 + 1 - x^2}{x - 1}$$

$$= \lim_{x \rightarrow 1} \left[\frac{\cancel{x-1}}{\cancel{x-1}} + \frac{1-x^2}{x-1} \right]$$

$$= \lim_{x \rightarrow 1} \left[1 + \frac{(1-x)(1+x)}{x-1} \right]$$

$$= \lim_{x \rightarrow 1} \left[1 + \frac{\cancel{-(x-1)}(1+x)}{\cancel{x-1}} \right] = 1 - 2 = -1$$

$\Rightarrow \therefore$ the slope of the tangent line $= -1$

(b) Equation of the tangent line:

$$y - y_0 = m(x - x_0)$$

$$y - 0 = -1(x - 1)$$



Text book questions



Find an equation of the tangent line to the curve at the given point

(Use definition) $y = x^3 - 3x + 1$, $(2, 3)$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$f'(2) = \lim_{x \rightarrow 2} \frac{x^3 - 3x + 1 - [8 - 6 + 1]}{x - 2}$$

$$= \lim_{x \rightarrow 2} \frac{x^3 - 3x + 1 - 8 + 6 - 1}{x - 2}$$

$$= \lim_{x \rightarrow 2} \frac{x^3 - 8 - 3x + 6 + 1 - 1}{x - 2}$$

$$= \lim_{x \rightarrow 2} \left[\frac{x^3 - 8}{x - 2} + \frac{-3x + 6}{x - 2} \right]$$

$$= \lim_{x \rightarrow 2} \left[\frac{(x-2)(x^2+2x+4)}{x-2} + \frac{-3(x-2)}{x-2} \right]$$

$$= 4 + 4 + 4 - 3 = 9$$

∴ The equation of the tangent line :

$$y - y_0 = m(x - x_0)$$

$$y - (3) = 9(x - 2)$$



Text book questions



Find an equation of the tangent line to the graph of $y = g(x)$ at $x = 5$ if $g(5) = -3$ and $g'(5) = 4$

$$g'(5) = 4, \quad P(5, -3)$$

∴ Equation of the tangent Line :

$$y - y_0 = m(x - x_0)$$

$$y - (-3) = 4(x - 5)$$

If the tangent line to the $y = f(x)$ at $(4, 3)$ passes through the point $(0, 2)$, Find $f(4)$ and $f'(4)$.

$$y = f(x) \quad P(4, 3) \longrightarrow f(4) = 3$$

Equation of the tangent line

$$y - y_0 = m(x - x_0)$$

$$y - 3 = m(x - 4)$$

∴ the tangent line passes through the point $(0, 2)$

$$2 - 3 = m(0 - 4)$$

$$-1 = -4m \longrightarrow$$

$$m = \frac{1}{4}$$

$$\begin{aligned} \rightarrow f'(4) \\ = \frac{1}{4} \end{aligned}$$

∴ The Equation of the tangent line

$$y - 3 = \frac{1}{4}(x - 4) \longrightarrow y - 3 = \frac{1}{4}x - 1$$

$$y = \frac{1}{4}x + 2$$



Tricky question



If $f'(2) = -8$ then find

$$\lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x^3 - 8}$$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$f'(2) = \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2}$$

$$-8 = \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2}$$

$\Rightarrow$

$$\lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x^3 - 8}$$

$$= \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{(x-2)(x^2 + 2x + 4)}$$

$$= -8 \cdot \frac{1}{4 + 4 + 4}$$

$$= -8 \cdot \frac{1}{12} = -\frac{2}{3}$$

تھوڑا
مباشر



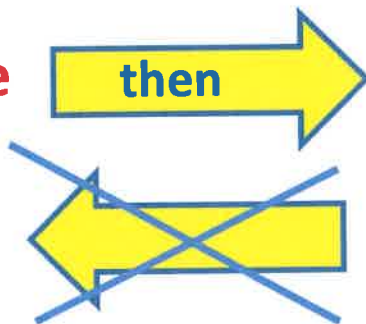
Differentiable at $x = a$



If f is Differentiable

then

f is Continuous.



العكس غير صحيح



f is differentiable at $x = a$ if

$$f'(\overset{+}{a}) = f'(\bar{a})$$

$$\lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a}$$

Use definition of derivative to show that f is differentiable at $x = -1$ where

$$f(x) = \begin{cases} -2x - 1, & x \leq -1 \\ x^2, & x > -1 \end{cases}$$

$$f'(a) = \lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a}$$

$$f'(1)^+ = \lim_{x \rightarrow 1^+} \frac{x^2 - 1}{x - (-1)}$$

$$= \lim_{x \rightarrow 1^+} \frac{(x-1)(x+1)}{x+1}$$

$$= -1 - 1$$

$$= -2$$

$$f'(a) = \lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a}$$

$$f'(-1)^- = \lim_{x \rightarrow -1^-} \frac{(-2x-1) - (-2-1)}{x - (-1)}$$

$$= \lim_{x \rightarrow -1^-} \frac{-2x - 1 - 1}{x + 1}$$

$$= \lim_{x \rightarrow -1^-} \frac{-2x - 2}{x + 1}$$

$$= \lim_{x \rightarrow -1^-} \frac{-2(x+1)}{x+1}$$

$$= -2$$

$\therefore$ at $x = -1 \rightarrow f$ is differentiable.

Find the constant A and B so that the function f will be differentiable at $x = -1$ where

$$f(x) = \begin{cases} Ax - 1, & x < -1 \\ 2x^2 + B, & x \geq -1 \end{cases}$$

$$f'(a) = \lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a}$$

$$f'(-1^+) = \lim_{x \rightarrow -1^+} \frac{(2x^2 + B) - (2 + B)}{x - (-1)}$$

$$= \lim_{x \rightarrow -1^+} \frac{2x^2 + B - 2 - B}{x + 1}$$

$$= \lim_{x \rightarrow -1^+} \frac{2(x^2 - 1)}{x + 1}$$

$$= \lim_{x \rightarrow -1^+} \frac{2(x-1)(x+1)}{x+1}$$

$$= 2(-2) = -4$$

$$f'(a) = \lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a}$$

$$f'(-1^-) = \lim_{x \rightarrow -1^-} \frac{(Ax - 1) - (-A - 1)}{x - (-1)}$$

$$= \lim_{x \rightarrow -1^-} \frac{Ax - 1 + A + 1}{x + 1}$$

$$= \lim_{x \rightarrow -1^-} \frac{A(x + 1)}{x + 1}$$

$$= A$$

$\therefore f$ is differentiable at $x = -1 \longrightarrow \therefore A = -4$
at $x = -1$

$$f(-1) = 2 + B$$

$$\lim_{x \rightarrow -1^+} (2x^2 + B) = 2 + B$$

$$\lim_{x \rightarrow -1^-} (Ax - 1) = -A - 1$$

$\therefore f$ is differentiable at $x = -1$

$\therefore f$ is continuous at $x = -1$

$$\therefore f(-1) = \lim_{x \rightarrow -1^+} f(x) = \lim_{x \rightarrow -1^-} f(x)$$

$$2 + B = 2 + B = -A - 1$$

$$2 + B = -A - 1$$

$$2 + B = -(-4) - 1$$

$$2 + B = 4 - 1$$

$$2 + B = 3 \longrightarrow B = 1$$



Harry up !!



Give an example for a function which is continuous but not differentiable at $x = 1$.

Past Exam

$$f(x) = |x - 1|$$

Is it true that if a function f is continuous at a , then f is differentiable at a ?

Past Exam

No, it is Not true.

$$f(x) = |x - 1|$$

f is continuous at $x=1$ but not differentiable at $x=1$

Give an example for a function which is continuous and differentiable at $x = 5$.

$$f(x) = x$$

....

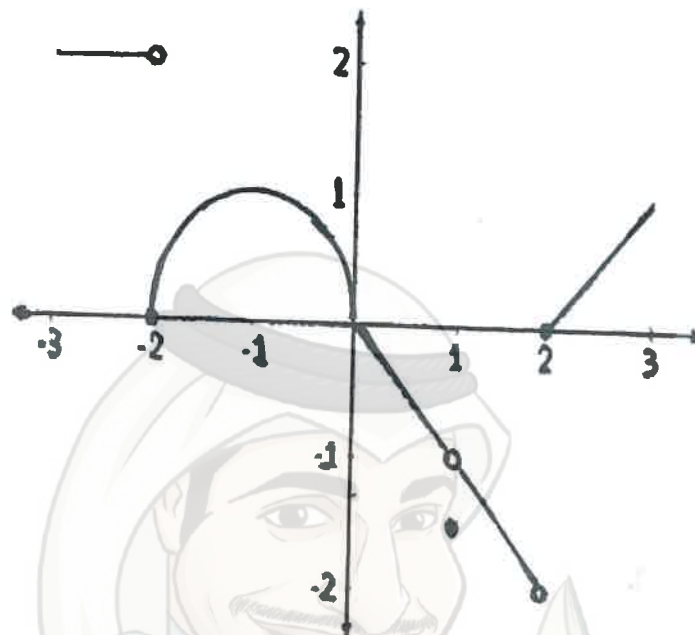
هذا اي صورية
عاريه

Consider the graph of $f(x)$ given in the bottom.

(a) Find all points, if any, at which f is discontinuous.

(b) Find all points, if any, at which f is non differentiable.

(c) Find $f'(-1)$



Past exam

(a) at $x=1 \rightarrow$ removable discontinuity
 $(f(1) \neq \lim_{x \rightarrow 1} f(x))$

يفرق الاتجاهين : 2 , -2
 يفرق الموضع : 1

at $x=2 \rightarrow$ Jump discontinuity

$$(\lim_{x \rightarrow 2^+} f(x) \neq \lim_{x \rightarrow 2^-} f(x))$$

at $x=-2 \rightarrow$ Jump discontinuity

$$(\lim_{x \rightarrow -2^+} f(x) \neq \lim_{x \rightarrow -2^-} f(x))$$

(b) f is not differentiable at

$x=1$, $x=2$, $x=-2$
 and $x=0$

(c) $f'(-1) = 0$